Real-Time Bluetooth Man-in-the-Middle Attack Detection System Based on Graph Neural Network

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Abstract—**With the popularization of Bluetooth technology, its security issues become especially critical. As a typical security threat, Bluetooth man-in-the-middle attack can cause serious damage to the integrity and privacy of data transmission. To address this challenge, this paper proposes a real-time Bluetooth signal anomaly detection model based on graph neural network (GNN). The model focuses on detecting and identifying potential man-in-the-middle attacks from complex Bluetooth signals. We first construct a Bluetooth signal dataset containing rich scenarios that model diverse communication environments and potential attack patterns. Using a graph representation learning approach, we convert the signals into graph structures, enabling the use of graph topology to analyze signal properties. Then, we design a novel graph convolution operation that captures spatio-temporal features in the signals and efficiently extracts patterns associated with attack behaviors. In the experimental section, we demonstrate the significant advantages of the proposed model in terms of detection accuracy, robustness and real-time performance by comparing it with existing methods. The model not only exhibits high accuracy in anomaly detection on the dataset, but also meets the demand of real-time monitoring in terms of processing speed. In addition, we also evaluate the generalization ability of the model and demonstrate its stability and reliability on different environments and devices.**

Index Terms—**Bluetooth, Graph Neural Network, Man-in-the-Middle Attack, Real-Time Detection**

I. INTRODUCTION

A. Motivation and Problem Definition

Bluetooth technology, as a wireless communication method, plays an increasingly important role in the field of modern communication due to its convenience and wide range of application scenarios. However, with the proliferation of Bluetooth devices, their security issues have gradually been exposed, especially Man-in-the-Middle (MitM) attacks, which pose a serious threat to user privacy and data security by intercepting and tampering with communication data.

The process of a Man-in-the-Middle attack typically includes the following steps: First, the attacker intercepts the communication signals between Bluetooth devices; then, by analyzing and cracking the signal content, the attacker can obtain sensitive information; finally, the attacker may impersonate one party in the communication and send false information to the other party to carry out fraud or other malicious acts. Due to the openness of Bluetooth communication, this type of attack is relatively easy to implement and difficult for ordinary users to detect.

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B. Limitations of Prior Art

Current Bluetooth security measures are insufficient in effectively identifying and defending against man-in-the-middle attacks. These measures are typically based on device fingerprint authentication [1] and cryptographic authentication mechanisms [2].

Cryptographic methods rely on encryption algorithms and authentication processes to ensure data security, but they may be limited by high implementation complexity, significant computational costs, and potential negative impacts on user experience. Especially in rapidly changing network environments, these methods may struggle to adapt to emerging security threats.

Device fingerprint authentication methods, which analyze the characteristics of Bluetooth signals to identify device identities, can be further divided into traditional feature-based detection methods [3] and modern machine learning-based detection methods [4]. Feature-based methods construct classification models by extracting statistical features from signals but are susceptible to environmental changes and signal fluctuations, thereby reducing detection accuracy. Machine learning-based methods, while promising for complex pattern recognition, require extensive labeled data for training and may lack real-time performance, making it difficult to meet the demands of rapid detection.

Therefore, designing a mechanism that can accurately and swiftly detect man-in-the-middle attacks in Bluetooth signals, while overcoming the shortcomings of existing methods, has become a significant challenge in current research. This requires new methods to not only have high accuracy and real-time capabilities but also to exhibit good adaptability and generalization.

C. Framework in a Nutshell

The real-time Bluetooth Man-in-the-Middle attack detection system framework proposed in this study aims to comprehensively analyze Bluetooth signals and effectively identify potential attack behaviors through a series of innovative methods. The system framework can be summarized into four main components: data preprocessing, signal embedding in high-dimensional space, graph construction, and feature aggregation.

Firstly, the data preprocessing module is responsible for denoising, interpolating, and normalizing the collected RSSI signals to enhance signal quality and prepare for subsequent analysis. Following this, the signal embedding module transforms the preprocessed time series data into a high-dimensional feature space, extracting key signal features through polynomial feature construction and channel attention mechanisms.

Subsequently, the graph construction module utilizes Manhattan distance and the TOPK mechanism to build a graph structure from high-dimensional embedding vectors, laying the foundation for the node classification problem. Lastly, the feature aggregation module employs the GMMConv algorithm from graph neural networks to aggregate features and classify nodes within the graph, identifying potential Bluetooth Man-in-the-Middle attacks.

D. Challenges and Solutions

In building our Bluetooth man-in-the-middle attack detection system, we faced the following major challenges and proposed solutions accordingly: **Challenge 1: Modeling the Dynamic Bluetooth Environment** The dynamic nature of the Bluetooth environment, such as changes in device distribution and motion, creates challenges for attack detection systems. Traditional methods are difficult to adapt to this dynamism and identify anomalous behaviors effectively. To address this challenge, our system employs a dynamic graph construction method that uses a learnable adjacency matrix in a graph neural network (GNN) to adaptively model the spatial and temporal relationships between tags, ensuring that the graph accurately reflects the current state of the Bluetooth environment. **Challenge 2: Labeling Issues for Wireless Signals** Accurate labeling of wireless data is difficult in the field of wireless technology, especially in large-scale deployments. To overcome the limitations of manual labeling, we transform the detection problem into a semi-supervised learning task on graphs, which utilizes the graph structure and the inherent relationships between nodes to reduce the need for labeled samples and lower the labeling burden while maintaining high detection accuracy. **Challenge 3: Model construction in unbalanced datasets** The small number of anomalous labels relative to the number of normal labels in the Bluetooth dataset leads to the sample imbalance problem, which may affect the performance of machine learning models. To address the class imbalance problem, our framework employs GMMConv, which enhances the model's sensitivity to a small number of classes by selectively aggregating neighboring node features to improve the detection accuracy of rare events (e.g., man-in-the-middle attacks).

With these solutions, our system is able to detect anomalous behaviors in Bluetooth signals more accurately and shows good generalization ability and stability across different environments and devices.

E. Section Organization of this Paper

The remainder of this paper is organized as follows: **Section 2 (Related Work)** reviews the literature pertinent to this study, including the modeling of signal correlation, developments in graph neural networks, and their application in multivariate time series analysis. This section provides the theoretical and technical context for our proposed system. **Section 3 (Detailed Design of the System)** presents a comprehensive overview of the design and construction blocks of our Bluetooth Man-in-the-Middle attack detection system. We begin with a high-level architecture overview and then explain each building block in detail sequentially. **Section 4 (IMPLEMENTATION AND EVALUATION)** demonstrates the performance evaluation of the graph neural network-based framework proposed in this paper for Bluetooth Man-in-the-Middle attack detection. This section includes the construction of datasets, baseline comparisons, evaluation metrics, and the results of controlled experiments. **Section 5 (Conclusion)** concludes the paper by summarizing the contributions and discussing how our framework provides an innovative solution for detecting Bluetooth Man-in-the-Middle attacks. **References** lists all the literature cited throughout this study for readers interested in further in-depth exploration of specific topics or technologies.

II. RELATED WORK

Related work can be mainly categorized into cryptography-based man-in-the-middle attack detection and fingerprint-based man-in-the-middle attack detection.

A. Cryptography-based man-in-the-middle attack detection

Cryptography-based methods focus on protecting the communication process through encryption techniques to prevent attackers from intercepting and tampering with data. The basic process of such methods usually includes steps such as key exchange, data encryption, and authentication.

G. Nath Nayak and S. G. Samaddar et al. [5]provide insights into understanding the different variants of man-in-the-middle attacks by examining different man-in-the-middle attack styles, consequences, and possible solutions. S. M. Bellovin and M. Merritt [6] proposed a cryptographic-based protocol that resists dictionary attacks and enhances the security of cryptographic key exchange. K. M. Haataja and K. Hypponen [7] have comparatively analyzed the Bluetooth man-in-the-middle attack and proposed a new attack and corresponding countermeasures. J. Clark and P. C. van Oorschot [8] revisit the security of SSL and HTTPS, evaluating enhancements to the certificate trust model.

However, cryptography-based approaches may face challenges such as high implementation complexity, high computational cost and impact on user experience.

B. Fingerprint-based man-in-the-middle attack detection

Fingerprinting-based methods, on the other hand, identify attacks by analyzing the physical characteristics of some devices such as network traffic. The basic process of such methods includes steps such as traffic capture, feature extraction, and pattern recognition using machine learning algorithms.

Jerry John Kponyo, Justice Owusu Agyemang, and Griffith Selorm Klogo [9] proposed an endpoint man-in-the-middle attack detection technique based on Address Resolution Protocol (ARP) analysis by combining signal processing and machine learning techniques that achieved 99.72 in detection accuracy. Another researcher [10] explored the possibility of using machine learning models to detect endpoint man-in-the-middle attacks, demonstrating the effectiveness of the approach by applying machine learning models on ARP analysis. José Aveleira-Mata et al [11]. proposed a novel intelligent approach for detecting man-in-the-middle attacks in IoT systems based on a hybrid process that first implements feature extraction and then applies different supervised classification techniques. Other researchers [12] have used deep learning algorithms to detect man-in-the-middle attacks, emphasizing the importance of feature selection in training and testing data packets.

However, such methods may be limited by the accuracy of feature selection and the generalization ability of machine learning models.

III. DETAILED DESIGN OF THE SYSTEM

In this section, we first present the overview of the system, and then explain the building blocks in detail sequentially.

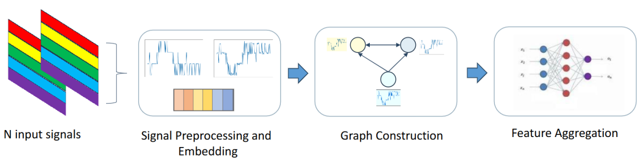

Fig. 1. System Flowchart

A. System Overview

We select and improve the traditional fingerprint authentication-based man-in-the-middle attack detection algorithm and use RSSI (Received Signal Strength Indicator) as the key physical fingerprint. RSSI is chosen as the signal fingerprint mainly because it can intuitively reflect the signal strength and distance information of Bluetooth devices, which is a unique advantage in the field of Bluetooth security.RSSI's ease of collection and sensitivity to distance variations make it an ideal indicator for monitoring.

First, RSSI is simple to measure and the data is easy to obtain because most Bluetooth devices are capable of providing RSSI values. Second, RSSI is able to reflect the distance and relative position information between devices. In normal Bluetooth communication, the RSSI value varies with the distance between devices. If an attacker intervenes in the normal communication, due to its positional difference with the normal communicating devices, it will produce a significant change in the RSSI signal, thus showing an abnormal pattern.

However, RSSI values are also susceptible to environmental factors such as obstacles, electromagnetic interference, and device differences, which can introduce noise and instability. These factors may lead to false alarms or missed alarms, which is a critical issue that needs to be addressed in our system. In order to improve the accuracy and robustness of RSSI-based man-in-the-middle attack detection, our system employs advanced signal processing techniques and machine learning algorithms to recognize and adapt to normal variations and abnormal patterns in RSSI signals.

With this approach, we aim to fully utilize the advantages of RSSI while reducing the adverse effects of environmental factors and equipment differences to improve the reliability and effectiveness of the detection system.

As illustrated in figure1, our system is designed to efficiently detect Bluetooth Man-in-the-Middle attacks and consists of four main components: Preprocessing, Signal Embedding in High-Dimensional Space, Graph Construction, and Feature Aggregation. Below is a brief introduction to these four parts.

1) **Preprocessing**: This part is responsible for denoising, interpolating, and normalizing the collected RSSI signals to improve signal quality and prepare for subsequent analysis.
2) **Signal Embedding in High-Dimensional Space**: In this stage, we transform the original time series data into a high-dimensional feature space, using polynomial feature construction and channel attention mechanisms to extract key features of the signal.
3) **Graph Construction**: Utilizing Manhattan distance and the TOPK mechanism, we construct a graph structure from the high-dimensional embedding vectors, laying the foundation for the node classification problem.
4) **Feature Aggregation**: Finally, we use the GMMConv algorithm from graph neural networks to aggregate features and classify nodes in the graph, identifying potential Bluetooth Man-in-the-Middle attacks.

Through these four steps, our system can comprehensively analyze Bluetooth signals and effectively identify and defend against Man-in-the-Middle attacks.

B. Preprocessing

In this paper, we first collect RSSI signals from various Bluetooth devices in different environments. After collection, we segment the continuous RSSI signals into multiple time windows, each capturing a snapshot of the signal. The detailed preprocessing steps are as follows:

Firstly, we perform spline interpolation on the signals within each time window to fill in any potential data gaps and smooth the noise. The mathematical expression for spline interpolation is given by:

$$S(t) = \text{spline\_interpolation}(\text{RSSI\_signal}) \tag{1}$$

Here, t represents the time point, and $S(t)$ represents the signal after interpolation.

Next, we normalize the interpolated signals to map their value ranges to the [0, 1] interval, eliminating the impact of different signal scales and value ranges, thereby enhancing the model's generalization ability. The normalization formula is defined as:

$$x_{\text{norm}} = \frac{x - \min(x)}{\max(x) - \min(x)} \tag{2}$$

In this formula, x is the original signal value, and x_{norm} is the normalized signal value.

On the basis of normalization, we further proceed with feature engineering. Using polynomial feature construction methods, we transform the original time series $y(t)$ into a higher-dimensional time series, such as $\{y(t), y^2(t), y^3(t), \ldots\}$, to increase the data's expressive power.

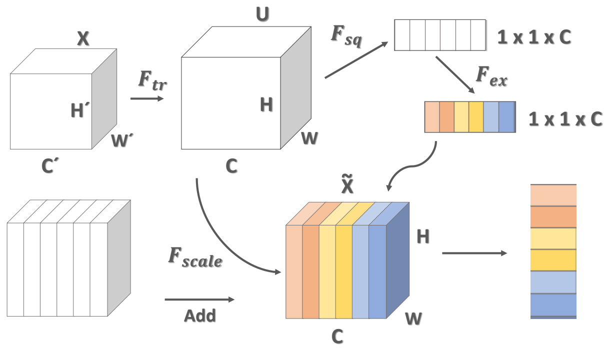

Fig. 2. Channel Attention Mechanisms

C. Signal Embedding in High-Dimensional Space

This section provides a detailed introduction of how to use the channel attention mechanism [13] to embed the features of Bluetooth RSSI signals, which have undergone feature engineering, into a high-dimensional space. Initially, as illustrated in figure2, we perform Global Average Pooling (GAP) on the input feature map $F \in \mathbb{R}^{H \times W \times C}$ to obtain the global spatial information of each channel, represented by the equation:

$$\text{GAP}(F_c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} F_c(i, j) \tag{3}$$

Here, F_c denotes the feature map of the c-th channel, where H and W correspond to the height and width of the feature map, respectively.

Subsequently, we reorganize the features through a fully connected layer (FC) following the GAP results and introduce non-linearity by using an activation function to transform and enhance the features:

$$\text{FC}(\text{GAP}(F_c)) = \mathbf{W} \cdot \text{GAP}(F_c) + b \tag{4}$$

Then, we generate gating weights using the Sigmoid activation function and adaptively adjust the original feature map through element-wise multiplication (Hadamard product):

$$\text{SA}(F_c) = F_c \odot \sigma(\text{FC}(\text{GAP}(F_c))) \tag{5}$$

In this equation, σ represents the Sigmoid function, and $\odot$ denotes the element-wise multiplication.

Through the channel attention mechanism, we obtain the weighted feature map $\text{SA}(F_c)$, which is then fed into a feedforward neural network (FNN) to learn the embedding representation of the signal in high-dimensional space. The embedding representation by the FNN can be expressed as:

$$\text{Embedding} = \text{FNN}(\text{SA}(F)) \tag{6}$$

The resulting embedding vector Embedding represents the signal in the high-dimensional space, providing rich feature information for further analysis and classification tasks.

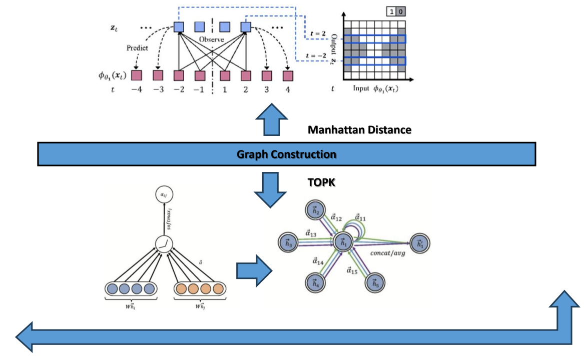

Fig. 3. Graph Construction Process

D. Graph Construction

In Bluetooth signal analysis, especially within complex networks with numerous devices and dynamically changing environments, traditional data analysis methods often struggle to effectively capture the complex interactions and relationships between devices. In contrast, graph representation, with its unique capabilities, can naturally capture the complexity of relationships such as proximity and connectivity between devices. It integrates multidimensional signal features to provide a comprehensive analysis perspective. The flexibility of graph structures allows them to adapt to dynamic changes in Bluetooth signals, such as device movement or fluctuations in signal strength. Moreover, graph theory and graph machine learning algorithms provide strong analytical tools, and the visualization of graphs greatly enhances the interpretability of results. In this study, we adopt graph construction as a key step, transforming the high-dimensional embedding vectors of Bluetooth RSSI signals into a graph representation, providing a structured form of data for signal analysis and man-in-the-middle attack detection.

As illustrated in figure3, we first define the nodes in the graph, each representing a Bluetooth device or signal source. Then, we use Manhattan distance [14] as the measure of distance between nodes, defined by the following formula:

$$d_{\text{Manhattan}}(\mathbf{v}, \mathbf{u}) = \sum_{i=1}^{n} |v_i - u_i| \tag{7}$$

Here, $\mathbf{v}$ and $\mathbf{u}$ are the embedding vectors of two nodes in high-dimensional space, and v_i and u_i are their coordinates in the i-th dimension.

Furthermore, we employ the TOPK mechanism [15] [16] to select the K nearest neighbors for each node, constructing a sparse yet expressive graph structure. This not only reduces computational complexity but also retains the most important relationships between nodes. The edge weights in the graph are directly determined by the Manhattan distance, denoted as w_{edge}, reflecting the similarity or proximity between nodes, calculated as:

$$w_{\text{edge}}(\mathbf{v}, \mathbf{u}) = d_{\text{Manhattan}}(\mathbf{v}, \mathbf{u}) \tag{8}$$

In this way, we construct a weighted graph where nodes represent Bluetooth devices, edges indicate associations between devices, and weights reflect the proximity of devices. This graph representation provides rich structured information for subsequent graph analysis and machine learning tasks, which is key to understanding and detecting potential man-in-the-middle attacks in Bluetooth signals.

E. Feature Aggregation

After we obtained the graph topology, we transformed the problem of Bluetooth signal analysis into a graph-based node classification issue and utilized Graph Neural Networks (GNNs) for feature aggregation of the signals. We selected the GMMConv algorithm [17] as the core of our GNN due to its unique advantages in handling non-Euclidean data. The GMMConv algorithm not only adapts to the dynamic characteristics of Bluetooth signals and the complex interactions between devices but also provides adaptability to non-Euclidean data, local feature learning, and end-to-end training capabilities.

Compared to traditional graph convolutional networks, the parameterized method of GMMConv offers greater flexibility and expressiveness and is capable of capturing subtle but critical differences within Bluetooth signals. This flexibility makes GMMConv particularly effective in detecting Bluetooth man-in-the-middle attacks, especially when dealing with signal variations caused by different devices and environments.

The following equation represents the feature update mechanism in a graph neural network, where the feature of a node is aggregated from its neighbors' features weighted by a function of their feature differences.

$$\mathbf{x}'_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \frac{1}{K} \sum_{k=1}^{K} \mathbf{w}_k(\mathbf{e}_{i,j}) \odot \mathbf{\Theta}_k \mathbf{x}_j, \tag{9}$$

Here, X_i is the updated feature vector of node i, $\mathcal{N}(i)$ is the set of neighbors of node i, e_{ij} is the feature difference between node i and its neighbor j, and $W_k(e_{ij})$ is the weight assigned to the feature from neighbor j to node i for the k-th kernel. The weight $W_k(e_{ij})$ is defined as:

$$\mathbf{w}_k(\mathbf{e}) = \exp\left(-\frac{1}{2}(\mathbf{e} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{e} - \mu_k)\right) \tag{10}$$

In this equation, μ_k represents the mean of the k-th kernel, and the weight is computed based on the exponential of the negative squared Euclidean distance between the feature difference e_{ij} and the kernel mean μ_k, scaled by the kernel's variance. These parameters are learnable, allowing the network to adjust the feature aggregation process adaptively.

To further extract the temporal dependency features in the Bluetooth signals, we feed the output of the Graph Neural Network into the Long Short-Term Memory network (LSTM) [18]. LSTM can effectively capture the dynamic characteristics of time series data, which is crucial for understanding the evolution process of Bluetooth signals.

$$\text{Output} = \text{LSTM}(\text{GNN Input}) \tag{11}$$

By combining Graph Neural Networks and LSTM, our model can not only handle spatial relationships but also capture temporal dependencies, providing a comprehensive feature learning framework for Bluetooth signal analysis.

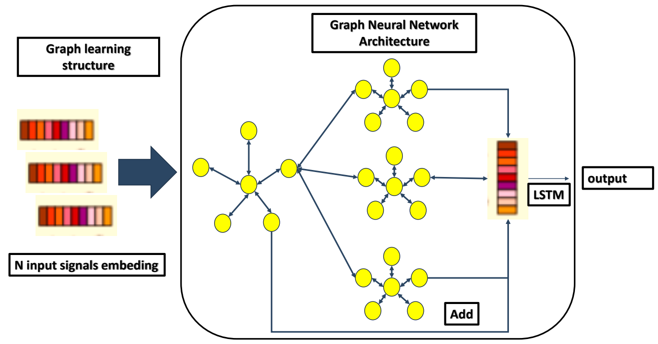

Fig. 4. Neural Network Architecture

In this study, we have designed an integrated model for analyzing and processing Bluetooth signals. The construction of the model begins with the establishment of the graph learning layer, which is responsible for capturing the complex relationships between Bluetooth devices. Unlike traditional deep learning architectures, the expressive power of Graph Neural Networks (GNNs) declines significantly as the number of layers increases due to the over-smoothing phenomenon, even with the addition of residual connections [19]. This is because the graph learning module essentially aggregates information from the node and its neighbors in various ways, and repeated message aggregation operations eventually make the feature representations of all nodes on the graph indistinguishable. To alleviate this phenomenon, this paper adopts the idea of the Inception structure [20], enhancing the model's expressive power by increasing the width of the network horizontally rather than its depth. Specifically, we introduce multiple branches with independent weights that extract features at various levels on the graph in parallel, enriching the diversity of features. As illustrated in figure4, we input the constructed graph and node features into the graph learning module and further send the output features into three independent graph learning sub-modules. These sub-modules extract features at different scales and integrate these features through scale fusion techniques. The output features of each sub-module are connected with the residual features of the previous level to achieve multi-scale feature fusion.

IV. IMPLEMENTATION AND EVALUATION

In this section, we evaluate the performance of the graph neural network-based Bluetooth man-in-the-middle attack detection framework proposed in this paper. Our goal is to

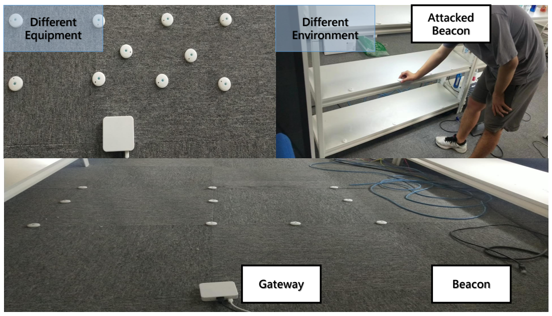

Fig. 5. Experimental Scene Diagram

demonstrate through a series of experiments that our framework has higher accuracy and greater robustness in detecting cloning attacks compared to traditional algorithms.

A. Experimental Setup

To thoroughly evaluate the performance of our proposed Bluetooth signal classification model based on Graph Neural Networks, we have constructed three Bluetooth datasets and designed a series of experiments. These experiments aim to test the model's accuracy, robustness, and generalization capabilities across different environments and devices.

As illustrated in figure5, we have carefully constructed three Bluetooth datasets as follows: The first dataset includes a certain number of Bluetooth devices, divided into training and testing sets for basic model training and performance evaluation; the second dataset is collected in different environments with the same devices as the first dataset, to verify the model's generalization ability in various settings; the third dataset is collected using different types of Bluetooth devices in the same environment as the first dataset, to verify the model's generalization ability across different devices.

In terms of hardware and software configuration, we have selected industry-recognized Bluetooth devices for data collection and developed the experimental software using the Python programming language. We utilized the PyTorch and PyTorch Geometric libraries to implement the graph neural network models and Scikit-learn for traditional classification algorithms. For model training and evaluation, we employed the cross-entropy loss function to guide the learning of the classification task and set the following hyperparameters: The number of iterations (Epochs) was set to 200 to ensure the model could fully learn, and the batch size (Batch Size) was set to 64 to achieve stable gradient updates and efficient memory usage. The dataset division ratio followed the standard 80/20 split, with the training set used for model learning and the testing set for model performance evaluation. Additionally, 5-fold cross-validation was applied to assess the model's stability and generalization ability. The optimizer selected was Adam, with an initial learning rate set to 0.001 and a decay strategy employed during training. Through these settings, we aim to ensure the rigor of the experiments and the comprehensiveness of the model evaluation.

B. Baseline

This paper uses common methods in the field of wireless sensing and some methods applied to time series as baselines:

1) Clustering [21]: A classic localization algorithm in the field of wireless sensing, which applies the K-Nearest Neighbors (KNN) and signal strength to infer the location of an unknown signal from the location of a known signal.
2) DeepWalk [22]: An algorithm that uses random walks on graphs to learn latent representations of nodes, capturing the network's structure for tasks such as node classification and prediction.
3) Line [23]: An approach that models the similarity between target nodes by comparing the similarity of their first-order neighbor features, creating scalable representations of nodes in networks.
4) PageRank [24]: A well-known algorithm that assigns numerical weights to each node in a network, indicating its relative importance or status within the network.
5) LSTM [25]: The algorithm utilizes Long Short-Term Memory (LSTM) networks to detect target information.
6) CNN+FFT [26]: The algorithm converts signals into a two-dimensional spectrogram through discrete short-time Fourier transform and employs a Convolutional Neural Network (CNN) to address the problem.
7) Transformer [27]: The algorithm applies the transformer architecture to solve anomaly detection in time series.
8) GraphSage [14]: The algorithm constructs time series into a complete graph and uses a GraphSage to tackle anomaly detection in time series.
9) Gat [15]: The algorithm uses cosine similarity to mine graph structures in time series and employs a Gat to detect anomalies in time series.

The above constitutes the baselines for the controlled experiments in this paper. Many of the algorithms are not specifically designed for localization, but they share the same underlying architecture. We demonstrate the effectiveness of our proposed framework by comparing its underlying architecture with those of other algorithms.

C. Evaluation Metrics

In this study, we conduct supervised anomaly detection with the aim of identifying abnormal signals from Bluetooth signals. We treat signals as nodes on a graph and their anomaly status as a binary classification problem, thereby transforming the original issue into a node classification problem on the graph. Given that the number of abnormal data points in the dataset is much smaller than that of normal data points, we select the following evaluation metrics to comprehensively assess the performance of the algorithmic model:

Precision measures the proportion of actual positive samples among all samples predicted as positive, with the formula:

$$\text{PRE} = \frac{TP}{TP + FP} \tag{12}$$

(a) Dataset 1 ACC (b) Dataset 1 F1 (c) Dataset 1 PRE (d) Dataset 1 REC

(e) Dataset 2 ACC (f) Dataset 2 F1 (g) Dataset 2 PRE (h) Dataset 2 REC

(i) Dataset 3 ACC (j) Dataset 3 F1 (k) Dataset 3 PRE (l) Dataset 3 REC

Fig. 6. Performance Metrics of Bluetooth Signal Anomaly Detection Across Three Datasets

Here, TP denotes the number of true positive samples, and FP denotes the number of false positive samples. Recall indicates the proportion of samples predicted as positive among all actual positive samples, with the formula:

$$\text{REC} = \frac{TP}{TP + FN} \tag{13}$$

FN represents the number of false negative samples. The F1 score is the harmonic mean of precision and recall, used to measure the balance between precision and recall, with the formula:

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

Accuracy represents the proportion of samples correctly classified among all samples, with the formula:

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

TN denotes the number of true negative samples. Through these evaluation metrics, we can comprehensively assess the model's performance in the task of Bluetooth signal anomaly detection, especially under the condition of unbalanced sample distribution, these metrics are crucial for measuring the model's accuracy and robustness.

D. Controlled Experiment

As shown in Figure 6, our graph neural network-based Bluetooth signal anomaly detection model was extensively tested on three different datasets. The experimental results are as follows: On Dataset 1, our model demonstrated an extremely high accuracy rate (ACC), approaching 0.9, and the F1 score also reached a satisfactory high level. This indicates that the model has excellent accuracy and robustness in distinguishing between normal and abnormal signals. The high values of precision (PRE) and recall (REC) further confirm the model's effectiveness in the specific classification tasks. The experimental results for Dataset 2 showed a slight decline in model performance, which may be due to the differences in the characteristics of abnormal signals in Dataset 2 compared to Dataset 1, thus posing higher requirements for the model's generalization ability. Even so, the model's performance in terms of ACC and F1 score remained at a high level, showing good adaptability and stability. On Dataset 3, the model once again showed high performance similar to that on Dataset 1, proving the model's adaptability and stability for signals collected by different devices. The results of ACC, F1 score, PRE, and REC were almost indistinguishable from those of Dataset 1, further verifying the model's robustness in different environments. Overall, comparing the experimental results of the three datasets, our model has shown high accuracy and stability in the task of Bluetooth signal anomaly detection. Although there are fluctuations in performance across different datasets, the model's anomaly detection capabilities are still very strong overall.

V. CONCLUSION

In this paper, we propose a graph neural network-based Bluetooth signal anomaly detection model, and verify the model's ability to detect Bluetooth man-in-the-middle attacks in different environments through a series of experiments. The experimental results show that compared with the traditional algorithms, our model shows significant improvement in evaluation metrics such as accuracy, F1 score, precision, and recall, displaying high accuracy and robustness. In addition, the stable performance of the model on different environments and devices proves its good generalization ability. This study improves the computational efficiency of the model while maintaining high expressive power by adopting the strategy of horizontal scaling rather than increasing the depth of the network. Despite the positive results of this study, there is still room for further optimization, and future work will focus on optimizing the model structure, improving the detection speed, and exploring the potential of the model for a wider range of practical applications. This study provides new perspectives and methods for Bluetooth signal anomaly detection, which is of great theoretical and practical significance for enhancing the security of wireless communications.

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